IGCSE 0607 Practice Questions — E3.5 — Combined Questions & Markscheme

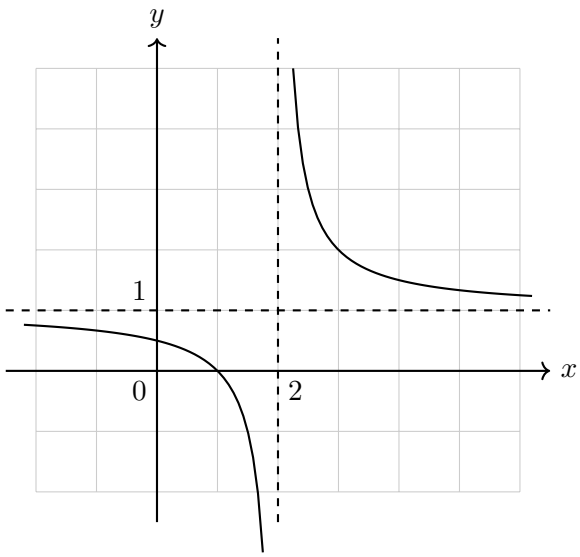
Asymptotes

Total Questions: 9 — **Total Marks:** 52 Marks

Table of Contents & Quick Navigation

- **E3.5** Asymptotes
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(5 Qs)*

1 The diagram shows the sketch of the graph of a function $y = f(x)$.
The dashed lines are the asymptotes to the curve.



(a) Write down the equation of the vertical asymptote.

..... **[1]**

(b) Write down the equation of the horizontal asymptote.

..... **[1]**

2 Let $f(x) = \dfrac{7}{x-4} + 3$.

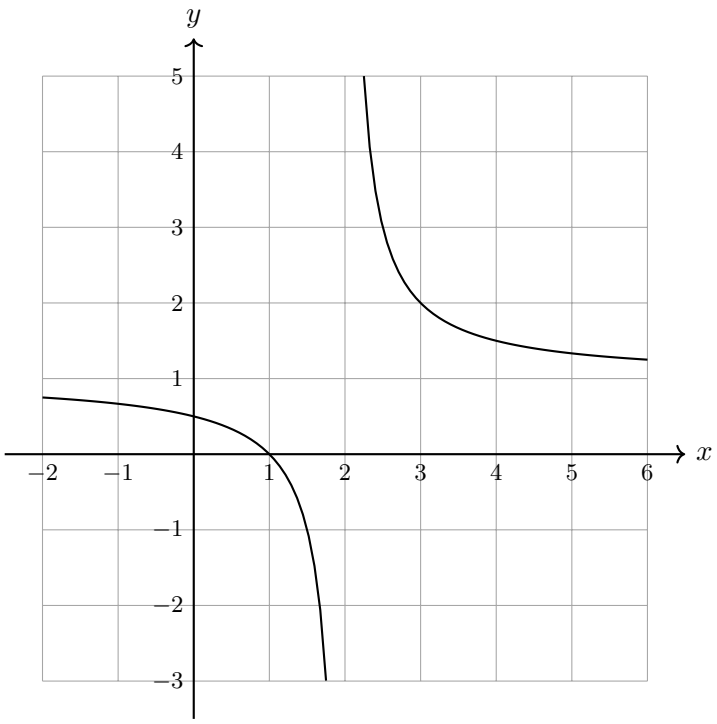
(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

3 The diagram shows the graph of a function $y = f(x)$.
The graph has two asymptotes, each parallel to one of the coordinate axes.
The equations of the asymptotes involve only integer values.



(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

(c) The graph of $y = f(x)$ is reflected in the y-axis to form the graph of $y = g(x)$.
Write down the equation of the vertical asymptote of the graph of $y = g(x)$.

..... **[1]**

4 A function is defined as $f(x) = \dfrac{4}{x-1} + 3$.

(a) Write down the equation of the

(i) vertical asymptote,

..... **[1]**

(ii) horizontal asymptote.

..... **[1]**

(b) Find the coordinates of the point where the graph of $y = f(x)$ crosses the x-axis.

((..... , ...)) **[2]**

(c) Find the coordinates of the point where the graph of $y = f(x)$ crosses the y-axis.

((..... , ...)) **[1]**

5 A function is defined as $f(x) = \dfrac{3}{x+2} + 1$.

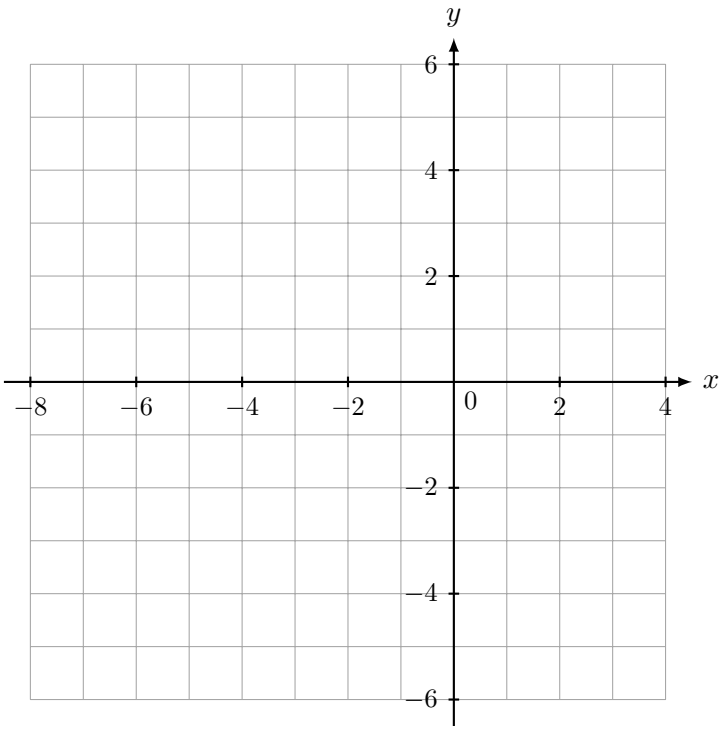
(a) Write down the equations of the two asymptotes of the graph of $y = f(x)$.

Vertical asymptote: ..... , Horizontal asymptote: ..... **[2]**

(b) Find the coordinates of the points where the graph of $y = f(x)$ crosses the x-axis and the y-axis.

x-axis at ..... , y-axis at ..... **[2]**

(c) On the grid, sketch the graph of $y = f(x)$.



[3]

6 Let $f(x) = \dfrac{3}{x+2} - 4$.

(a) Write down the equation of the vertical asymptote of the graph of $y = f(x)$.

..... **[1]**

(b) Write down the equation of the horizontal asymptote of the graph of $y = f(x)$.

..... **[1]**

(c) Find the coordinates of the points where the graph of $y = f(x)$ intersects the coordinate axes.

x-intercept = ... , y-intercept = ... **[5]**

(d) The graph of $y = \tan x$ has an infinite number of vertical asymptotes.
Write down the equations of the two asymptotes in the interval $0° \leq x \leq 360°$.

$x =$ ..... , $x =$ ..... **[2]**

7 A function is given by $f(x) = \dfrac{a}{x-b} + c$, where a, b and c are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 2$ and a horizontal asymptote at $y = -1$.
The graph intersects the x-axis at the point $(5, 0)$.

(a) Write down the value of b and the value of c.

$b =$ ..... , $c =$ ..... **[2]**

(b) Work out the value of a.

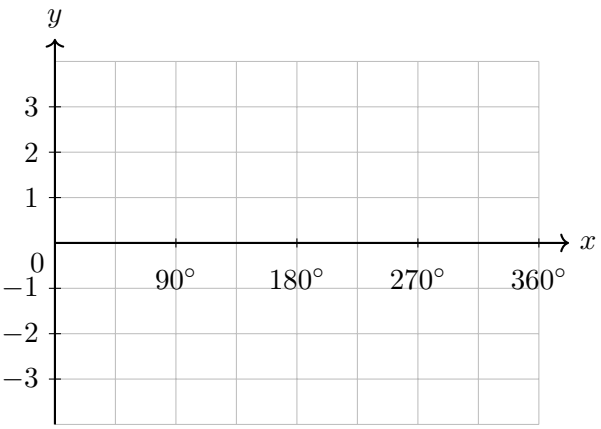
$a =$ ..... **[2]**

(c) The graph of $y = f(x)$ intersects the straight line $y = -x + 5$ at two points.
Find the x-coordinates of these two points.

$x =$ ..... , $x =$ ..... **[2]**

8 The function $g(x) = \tan x$ is defined for $0° \leq x \leq 360°$.

(a) On the grid, sketch the graph of $y = g(x)$.



[2]

(b) Write down the equations of the asymptotes of the graph of $y = g(x)$ in this domain.

[2]

(c) Solve the equation $2\tan x + 3 = 0$ for $0° \leq x \leq 360°$.
Give your answers correct to 1 decimal place.

$x =$ ..... , $x =$ ..... **[3]**

9 The function $f(x)$ is given by $f(x) = \dfrac{p}{x-q} + r$, where p, q, and r are constants.
The graph of $y = f(x)$ has a vertical asymptote at $x = 5$ and a horizontal asymptote at $y = -3$.
The graph passes through the point $(7, -2)$.

(a) Write down the value of q and the value of r.

$q =$ ..... , $r =$ ..... **[2]**

(b) Find the value of p.

$p =$ ..... **[3]**

(c) Find $f^{-1}(x)$.

$f^{-1}(x) =$ ..... **[4]**

(d) Write down the equations of the asymptotes of the graph of $y = f^{-1}(x)$.

$x =$ ..... , $y =$ ..... **[2]**

1 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

[Back to Question] *[Total Marks: 2]*

2 **(a)** $x = 4$ **[B1]**

(b) $y = 3$ **[B1]**

[Back to Question] *[Total Marks: 2]*

3 **(a)** $x = 2$ **[B1]**

(b) $y = 1$ **[B1]**

(c) $x = -2$ **[B1]**

[Back to Question] *[Total Marks: 3]*

4 **(a)** **(i)** $x = 1$ **[B1]**

(ii) $y = 3$ **[B1]**

(b) $\dfrac{4}{x-1} + 3 = 0$ **[M1]**

$(-\dfrac{1}{3}, 0)$ **[A1]**

(c) $(0, -1)$ **[B1]**

[Back to Question] *[Total Marks: 5]*

5 **(a)** Vertical asymptote: $x = -2$ **[B1]**

Horizontal asymptote: $y = 1$ **[B1]**

(b) Substitute $y = 0$ to find x-intercept: $0 = \dfrac{3}{x+2} + 1 \implies x = -5$ **[M1]**

x-axis at $(-5, 0)$ and y-axis at $(0, 2.5)$ **[A1]**

(c) Two correct branches of the hyperbola sketched **[B1]**

Graph approaches $x = -2$ and $y = 1$ correctly without touching or crossing them **[B1]**

Graph passes through $(-5, 0)$ and $(0, 2.5)$ approximately **[B1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** $x = -2$ **[B1]**

(b) $y = -4$ **[B1]**

(c) Sets $x = 0$ to find the y-intercept: $y = \dfrac{3}{0+2} - 4$ **[M1]**

y-intercept is $(0, -2.5)$ **[A1]**

Sets $y = 0$ to find the x-intercept: $0 = \dfrac{3}{x+2} - 4$ **[M1]**

$4 = \dfrac{3}{x+2} \implies 4(x+2) = 3 \implies 4x + 8 = 3$ **[M1]**

x-intercept is $(-1.25, 0)$ **[A1]**

(d) $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

[Back to Question] *[Total Marks: 9]*

7 (a) $b = 2$ **[B1]**

$c = -1$ **[B1]**

(b) Substitutes $(5, 0)$ into the equation: $\dfrac{a}{5-2} - 1 = 0$ **[M1]**

$a = 3$ **[A1]**

(c) Sets up the equation $\dfrac{3}{x-2} - 1 = -x + 5$ or uses GDC to find intersection of $y = f(x)$ and $y = -x + 5$ **[M1]**

$x = 3$ and $x = 5$ **[A1]**

[Back to Question] *[Total Marks: 6]*

8 (a) Two correct vertical asymptotes at $x = 90°$ and $x = 270°$ **[B1]**

Three correct branches passing through $(0, 0)$, $(180, 0)$, and $(360, 0)$ with correct general curvature **[B1]**

(b) $x = 90°$ **[B1]**

$x = 270°$ **[B1]**

(c) $\tan x = -1.5$ **[M1]**

$x = 123.7°$ **[A1]**

$x = 303.7°$ **[A1]**

[Back to Question] *[Total Marks: 7]*

9 (a) $q = 5$ **[B1]**

$r = -3$ **[B1]**

(b) Substitutes $x = 7, y = -2$ and their values of q and r into the equation: $-2 = \dfrac{p}{7-5} - 3$ **[M1]**

$-2 = \dfrac{p}{2} - 3 \implies \dfrac{p}{2} = 1$ **[M1]**

$p = 2$ **[A1]**

(c) Sets $y = \dfrac{2}{x-5} - 3$ and attempts to make x the subject, or swaps x and y **[M1]**

$y + 3 = \dfrac{2}{x-5} \implies x - 5 = \dfrac{2}{y+3}$ **[M1]**

$x = \dfrac{2}{y+3} + 5$ **[M1]**

$f^{-1}(x) = \dfrac{2}{x+3} + 5$ or equivalent **[A1]**

(d) $x = -3$ **[B1]**

$y = 5$ **[B1]**

(Strict follow through from their answers in part (a): $x = r, y = q$)

[Back to Question] *[Total Marks: 11]*